Chapter 5

# DNA SEQUENCING

I. General introduction

II. Maxam-Gilbert method

III. Sanger method

IV. NGS methods

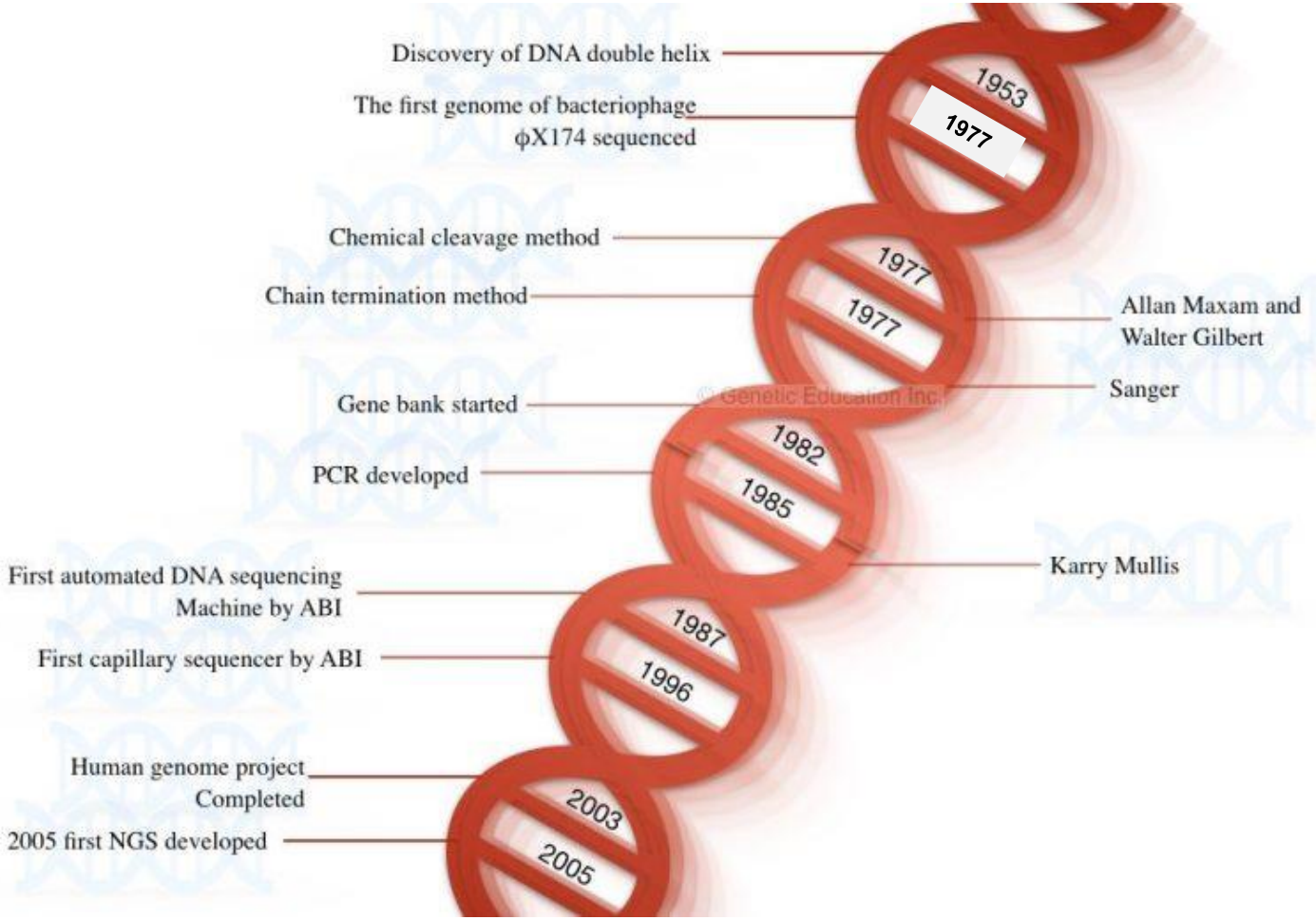
# I. General introduction

- **DNA sequencing** — determining the order of nucleotides (A, T, C, G) in a DNA genome/fragment.
- **Challenges**

  - Length of a DNA genome/fragment…

  - Four types of nu (A, T, G, C)

# Major milestones of DNA sequencing



# Application

- **Medicine and Health**

  *- disease diagnosis, pathogen identification,...*

- **Forensic science**
- **Agriculture**

  - Crop improvement:

  - Livestock genetics:

  - GMO characterization:

- **Microbiology and Environmental Studies**

  - Metagenomics: Study microbial communities in soil, oceans, animal gut,…

  - Epidemiology: Track disease outbreaks through pathogen genome sequencing.

- **Research and Biotechnology**

# Generations of sequencing

- **First generation sequencing**
  - Maxam-Gilbert method
  - Sanger method
- **Next generation sequencing (NGS)**

# II. Maxam-Gilbert method

*Proc. Natl. Acad. Sci. USA*
Vol. 74, No. 2, pp. 560–564, February 1977
Biochemistry



Nobel hóa học 1980

## A new method for sequencing DNA

(DNA chemistry/dimethyl sulfate cleavage/hydrazine/piperidine)

ALLAN M. MAXAM AND WALTER GILBERT

Department of Biochemistry and Molecular Biology, Harvard University, Cambridge, Massachusetts 02138

*Contributed by Walter Gilbert, December 9, 1976*

**ABSTRACT** DNA can be sequenced by a chemical procedure that breaks a terminally labeled DNA molecule partially at each repetition of a base. The lengths of the labeled fragments then identify the positions of that base. We describe reactions that cleave DNA preferentially at guanines, at adenines, at cytosines and thymines equally, and at cytosines alone. When the products of these four reactions are resolved by size, by elec-

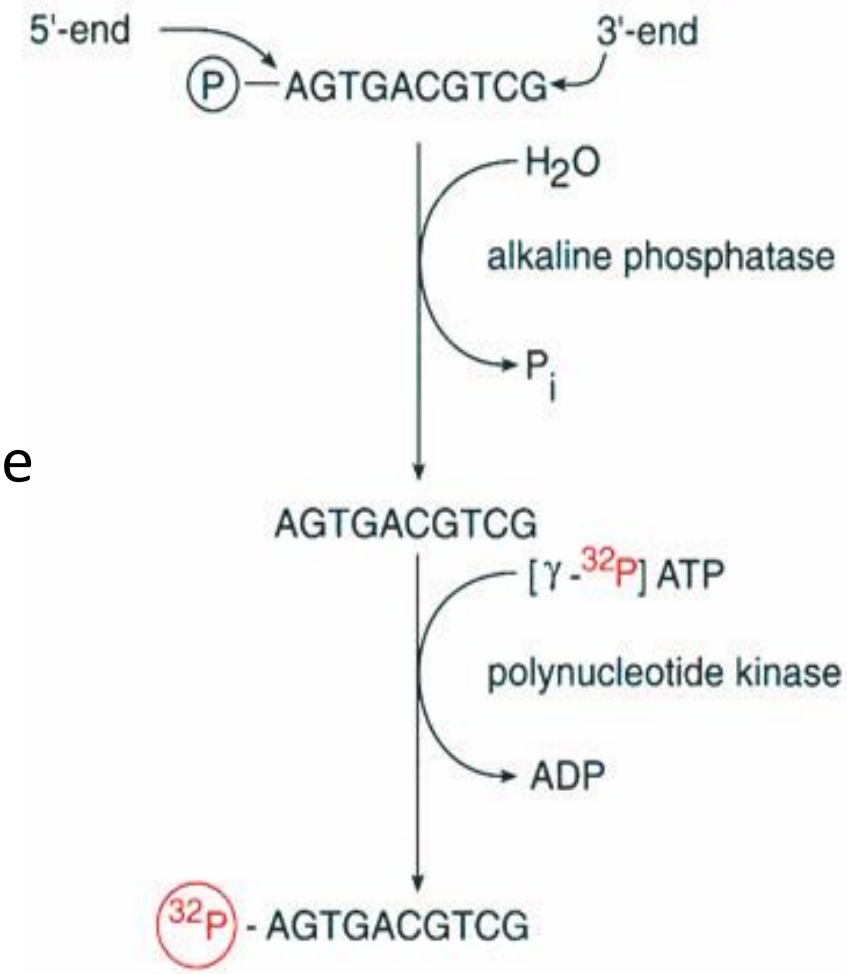
### THE SPECIFIC CHEMISTRY

**A Guanine/Adenine Cleavage (2).** Dimethyl sulfate methylates the guanines in DNA at the N7 position and the adenines at the N3 (3). The glycosidic bond of a methylated purine is unstable (3, 4) and breaks easily on heating at neutral pH, leaving the sugar free. Treatment with 0.1 M alkali at 90°

# Maxam-Gilbert method

- known as chemical DNA sequencing
- **Principle**

**Radioactive labeling** of one end of the DNA strand.

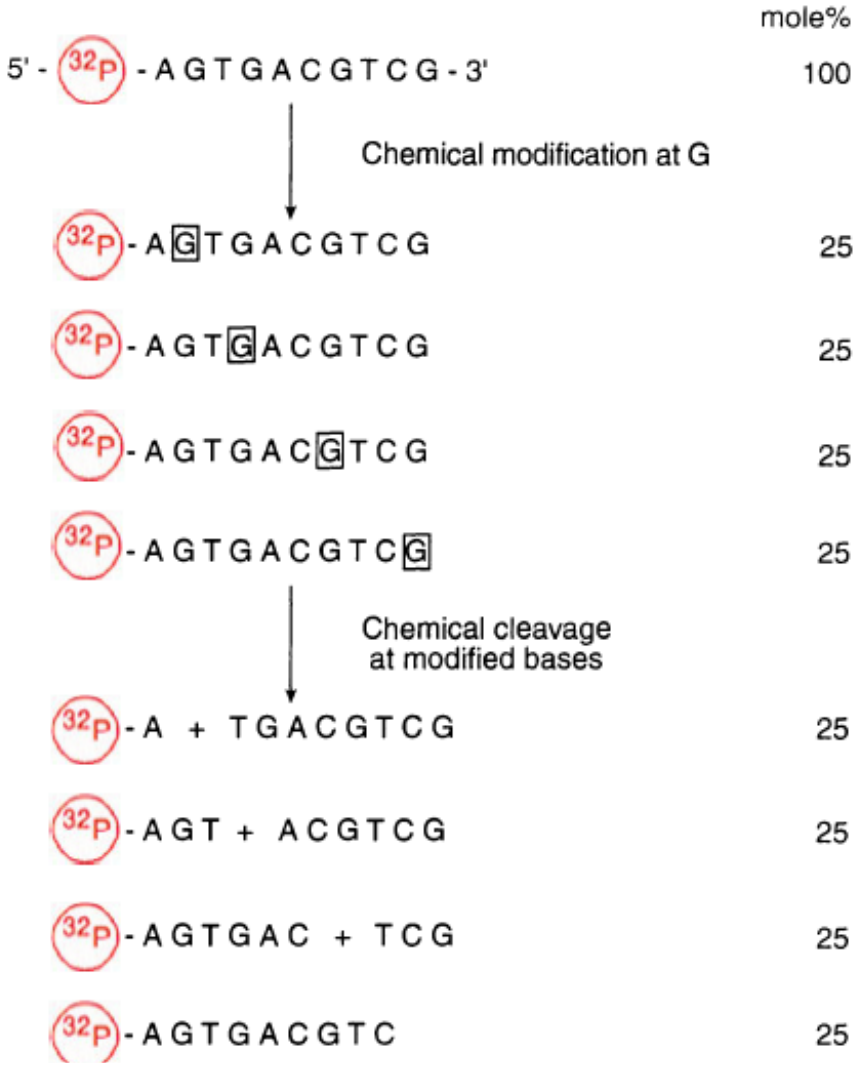


# Maxam-Gilbert method

- **Principle**

**Chemical treatment** to selectively modify specific bases.

**Chemical cleavage** of the DNA backbone at modified bases.

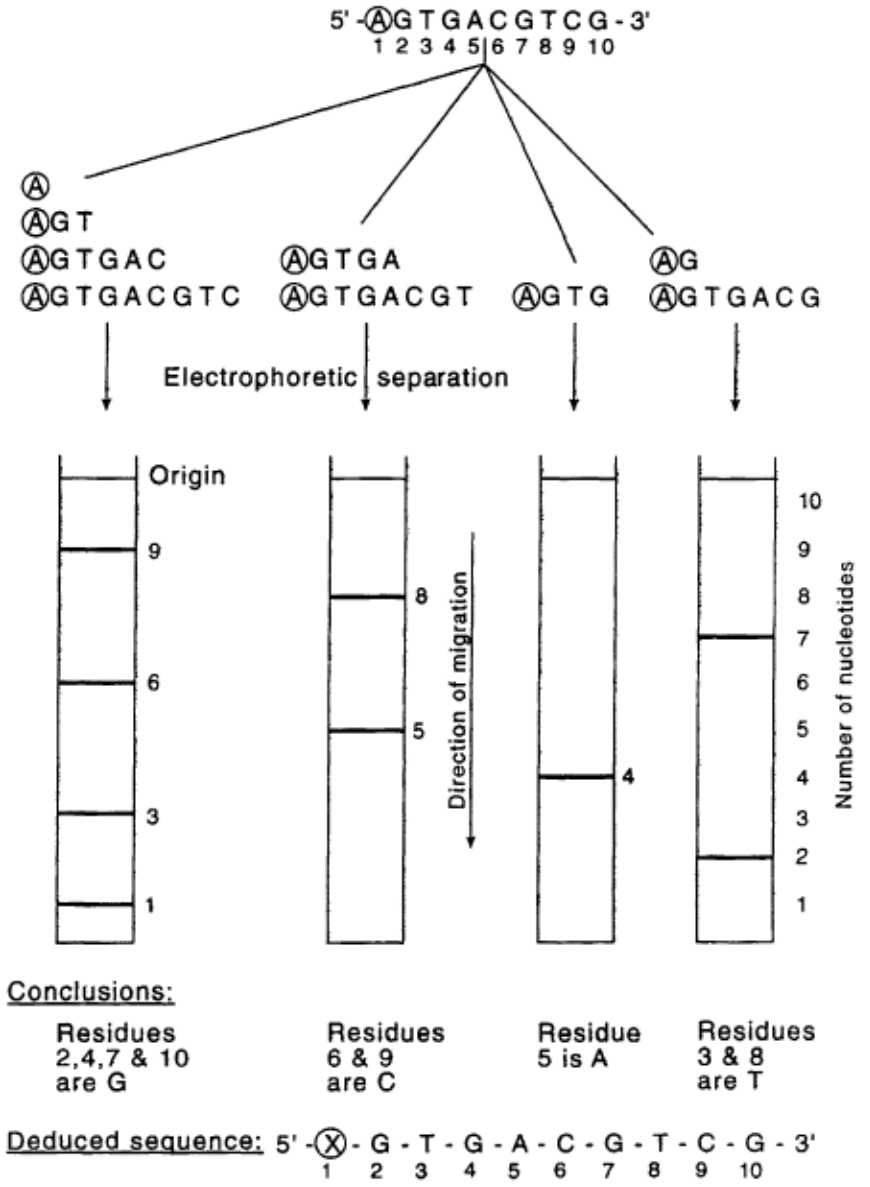


# Maxam-Gilbert method

- **Principle**

**Size separation** of fragments by denaturing polyacrylamide gel electrophoresis (PAGE).

**Film exposure** to determine fragment lengths and deduce the sequence.




(Biochemistry, Stenesh, 1998)

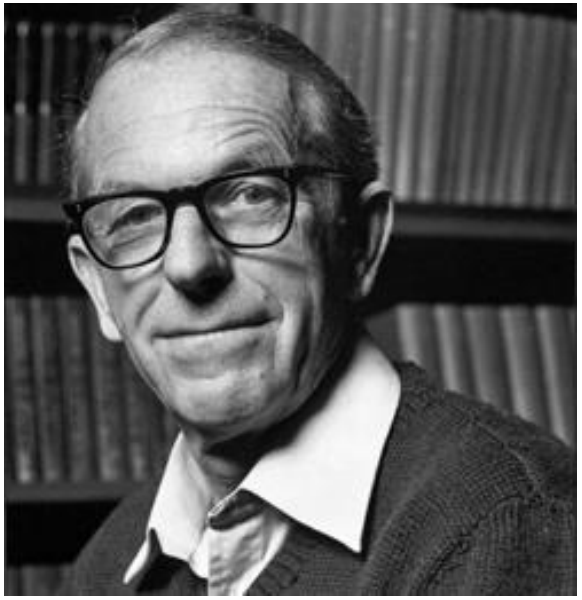
# Maxam-Gilbert method

- **Disadvantages**

  - Uses toxic and carcinogenic chemicals
  - Time-consuming and technically difficult
  - Poor scalability

# III. Sanger method

- By F. Sanger (1977)



*Proc. Natl. Acad. Sci. USA*
Vol. 74, No. 12, pp. 5463–5467, December 1977
Biochemistry

## DNA sequencing with chain-terminating inhibitors

(DNA polymerase/nucleotide sequences/bacteriophage $\phi$X174)

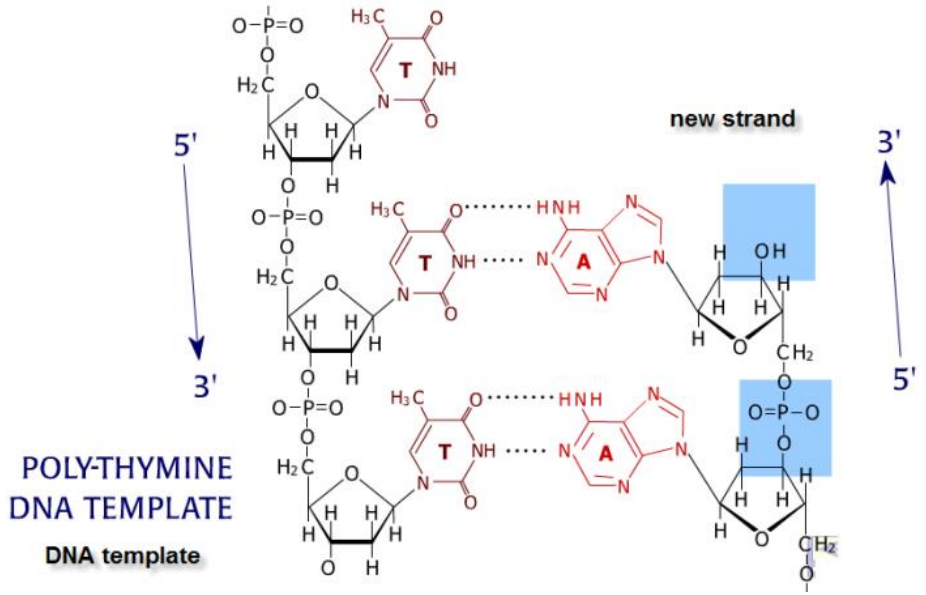
F. SANGER, S. NICKLEN, AND A. R. COULSON

Medical Research Council Laboratory of Molecular Biology, Cambridge CB2 2QH, England

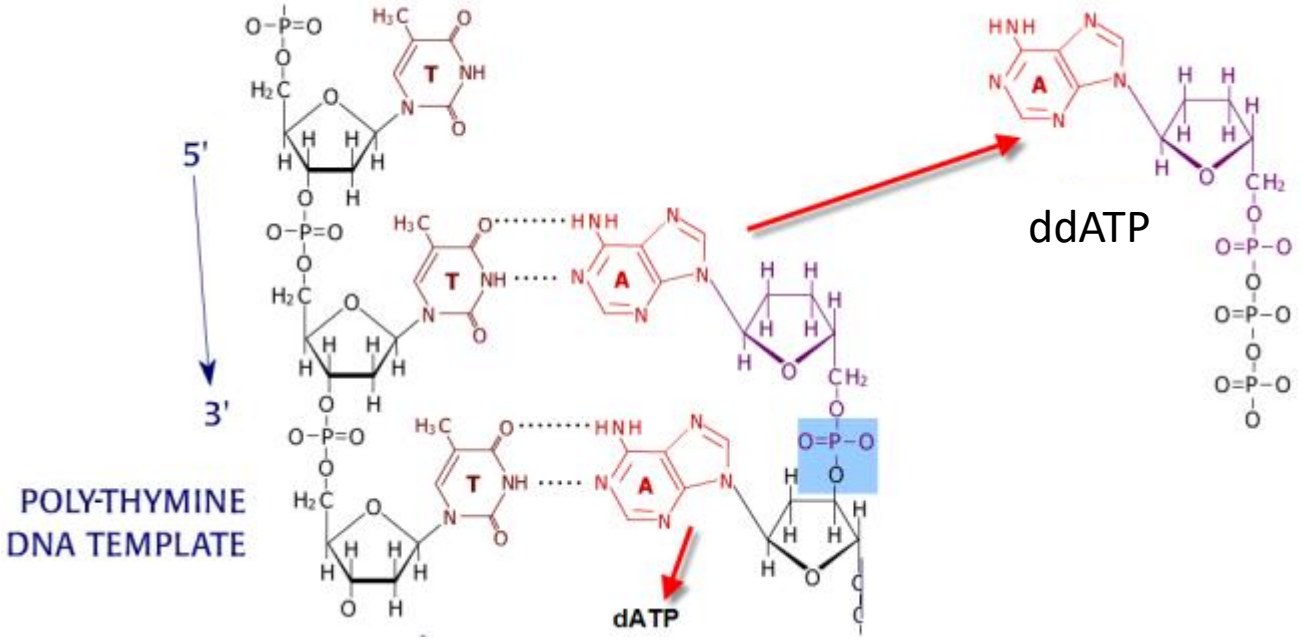
*Contributed by F. Sanger, October 3, 1977*

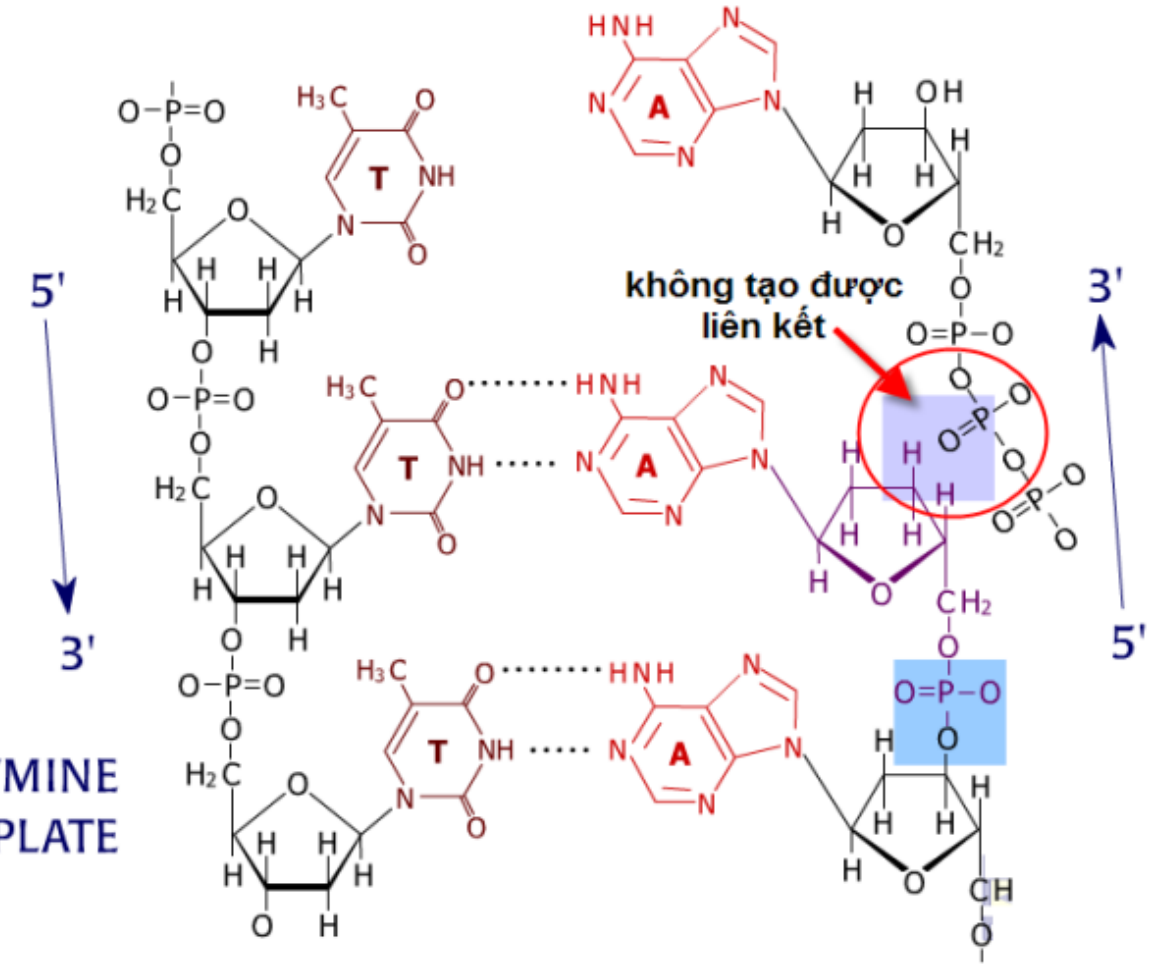
**ABSTRACT** A new method for determining nucleotide sequences in DNA is described. It is similar to the "plus and minus" method [Sanger, F. & Coulson, A. R. (1975) *J. Mol. Biol.* 94, 441–448] but makes use of the 2′,3′-dideoxy and arabinonucleoside analogues of the normal deoxynucleoside triphosphates, which act as specific chain-terminating inhibitors of DNA polymerase. The technique has been applied to the DNA of bacteriophage $\phi$X174 and is more rapid and more accurate than either the plus or the minus method.

a stereoisomer of ribose in which the 3′-hydroxyl group is oriented in *trans* position with respect to the 2′-hydroxyl group. The arabinosyl (ara) nucleotides act as chain terminating inhibitors of *Escherichia coli* DNA polymerase I in a manner comparable to ddT (4), although synthesized chains ending in 3′ araC can be further extended by some mammalian DNA polymerases (5). In order to obtain a suitable pattern of bands from which an extensive sequence can be read it is necessary

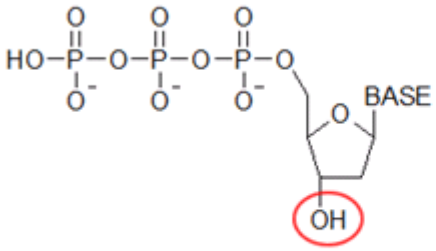



- **Principle**: use *"defective" DNA nucleotides ddNTPs* that lack the 3′–OH group

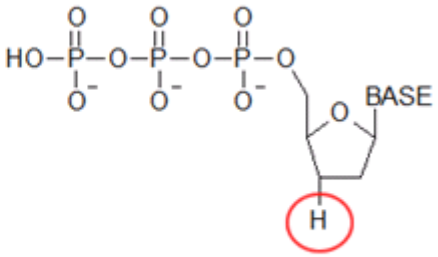





# Principle




deoxynucleotide triphosphate




dideoxynucleotide triphosphate

When a **ddNTP** is added to the growing chain:

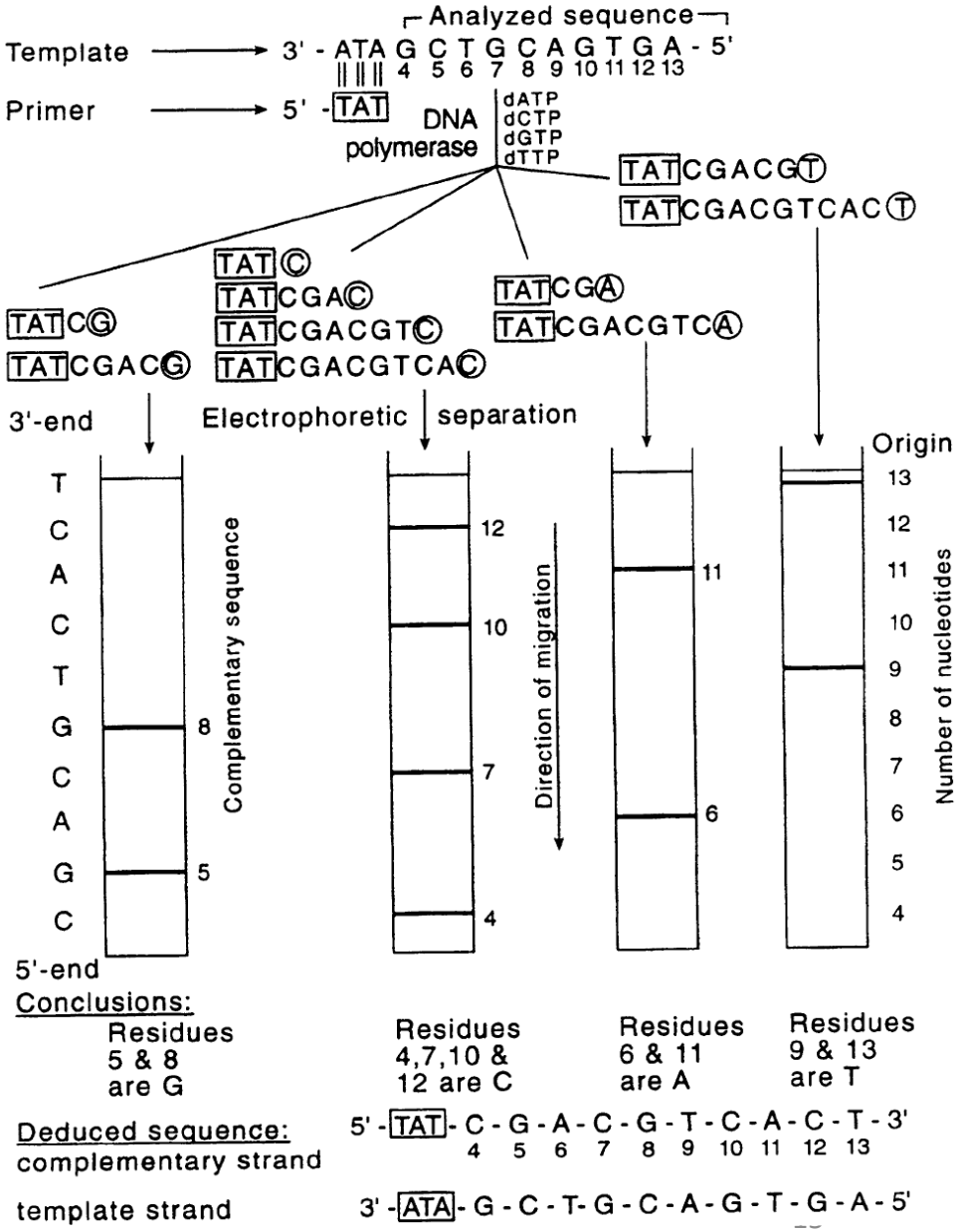
➞ The chain **cannot extend further**

➞ DNA synthesis **terminates**

# Protocol of 'classical' Sanger method

**Key Components**

- Template DNA (single-stranded)
- DNA primer
- DNA polymerase
- Mix of dNTPs
- Four ddNTPs (4 tubes), each tagged radioactive labels
- Separation by polyacrylamide gel electrophoresis → Autoradiography (X-ray film) of the gel (~Maxam-Gilber method)




(Biochemistry, Stenesh, 1998)

# ‘Classical’ Sanger method

- **Disadvantages**
  - Uses toxic and carcinogenic chemicals
  - Time-consuming and technically difficult
  - Poor scalability

# Automated Sanger method

- Use ddNTP fluorescent labels

- All ddNTPs are included in one tube, but each ddNTP carries a different fluorescent color.

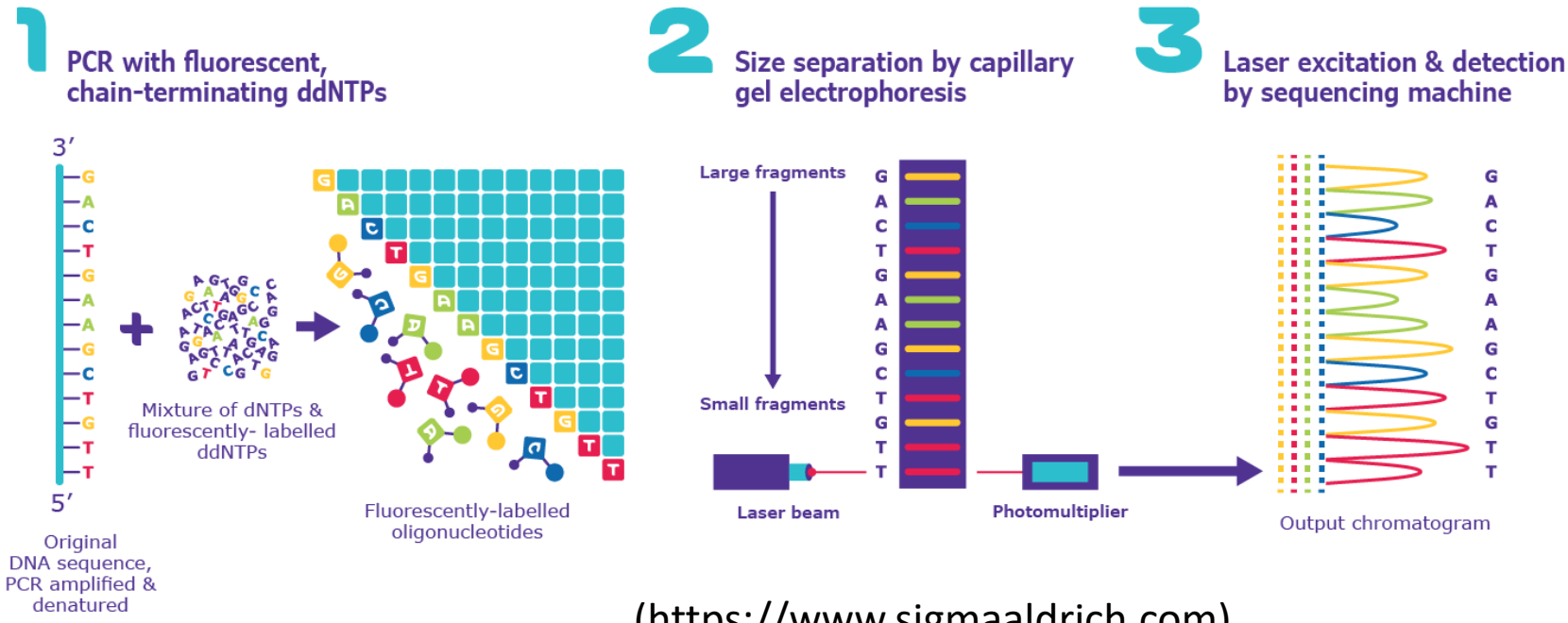
- Use capillary electrophoresis

- Use automated sequencing machine

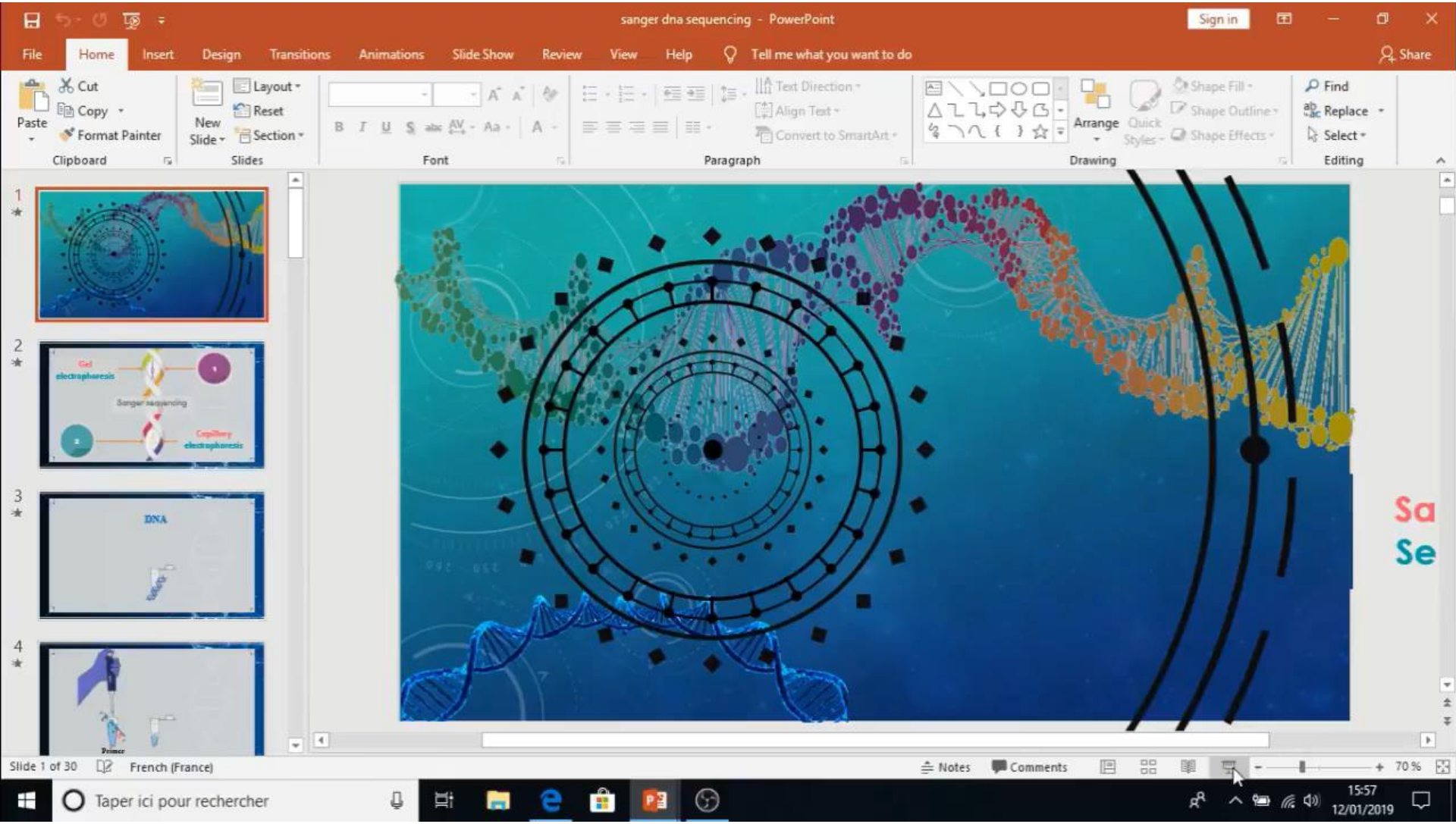
- Longer reads: 700 – 900 nu

**Advantages**

- ✓ High accuracy (99.9%)
- ✓ Long reads: 700 – 900 nu
- ✓ Gold-standard for confirming mutations
- ✓ Easy analysis
- ✓ Fast



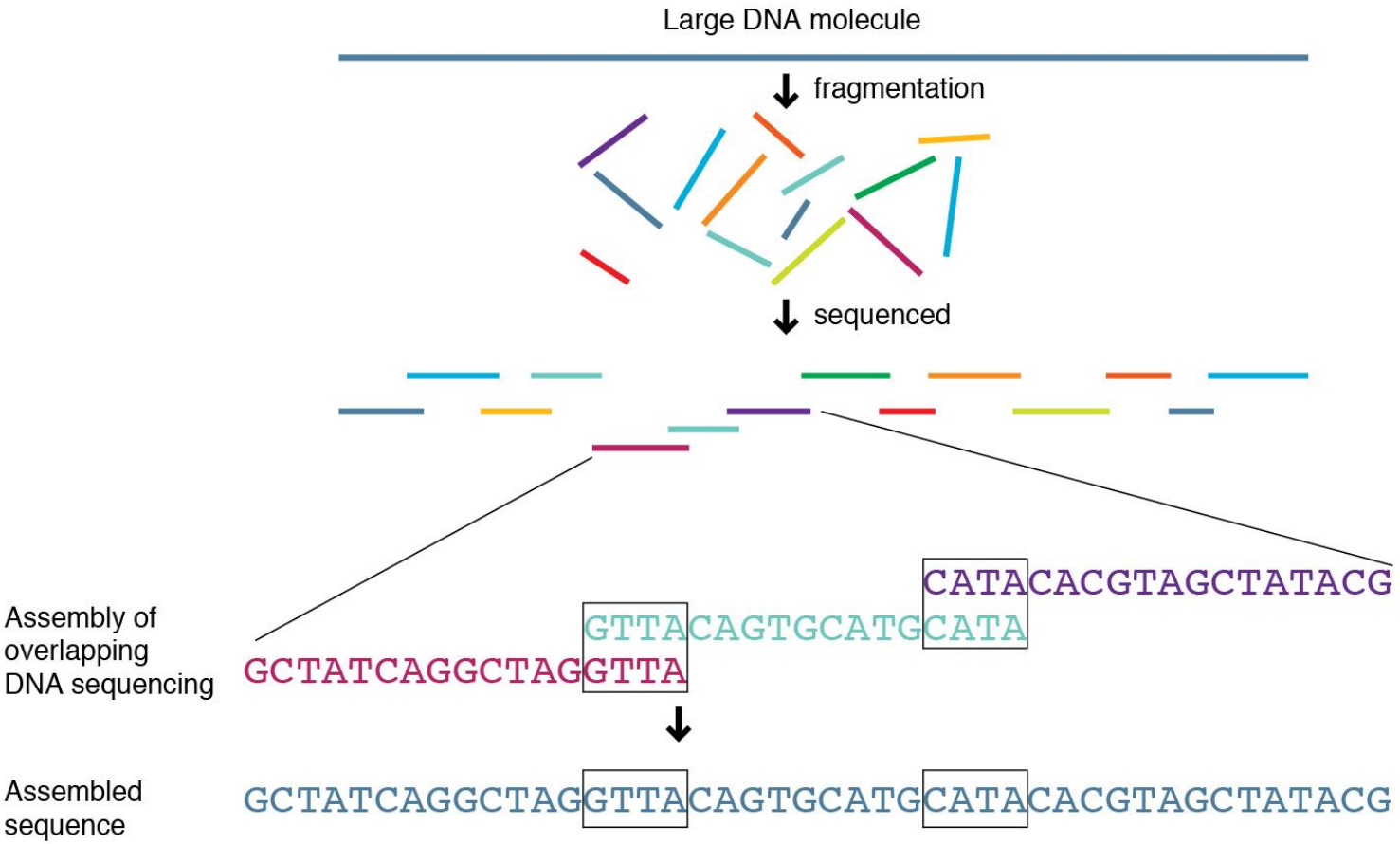

(https://www.sigmaaldrich.com)


sanger dna sequencing - PowerPoint
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Shape Outline
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1
2
Gel electrophoresis
Sanger sequencing
Capillary electrophoresis
3
DNA
4
Primer
Slide 1 of 30
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Notes
Comments
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# Dicussion

- Primer sequence?
- F – R primers!!!
- Sequencing a fragment ~3000 bp?

# Shot gun sequencing



# IV. Next-Generation Sequencing (NGS)

- Illumina (most dominant)
- Ion Torrent
- Oxford Nanopore
- PacBio SMRT

**Advantages of NGS**

✓ Extremely high throughput
✓ Low cost per base
✓ Multiplex many samples at once
✓ Detects rare variants
✓ Enables comprehensive genomic studies

**Disadvantages**

⚠ Requires complex bioinformatics
⚠ Large data storage (GB–TB)
⚠ Specialized equipment and expertise
⚠ Higher initial investment